

Smaller and Shallower XOR Circuits for AES MixColumns: 97 Gates at Depth 3; 92 at Depth 4; 89 at Depth 5

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Abstract

We give four verified implementations of AES MixColumns as circuits of 2-input XOR gates over $\text{GF}(2)$. Three improve the published depth–count Pareto frontier at their depth: **97** gates at depth **3**, the minimum depth, against 99 [2]; **92** at depth **4** against 97 [3]; **89** at depth **5** against 94 [3]. The fourth, **88** at depth **7**, *ties* the published gate-count floor of Jean [9] — who has priority — with an independent circuit sharing 61 of 88 internal masks. All were found by stochastic local search over a value-set formulation of the shortest-linear-program problem. No 87 was found: 47 canonical 88-gate circuits have exhaustively empty remove- ≤ 3 neighbourhoods. Nothing here is claimed optimal.

1 Model and specification

A circuit is an ordered list of 2-input XOR gates: $s_0, \dots, s_{31}$ are the input bits, and gate k produces $s_{32+k} = s_a \oplus s_b$ with $a, b < 32 + k$. Cost is the gate count; depth counts gates on the longest input-to-output path, inputs at depth 0. Bit index i denotes bit $(i \bmod 8)$ of state byte $\lfloor i/8 \rfloor$, least-significant-bit first, and MixColumns maps an input column $a_0, \dots, a_3$ to

$$\text{out}[c] = 2 \cdot a_c \oplus 3 \cdot a_{(c+1) \bmod 4} \oplus a_{(c+2) \bmod 4} \oplus a_{(c+3) \bmod 4},$$

in $\text{GF}(2^8)$ modulo $x^8 + x^4 + x^3 + x + 1$ (0x11b), per NIST FIPS 197-upd1, Eq. 5.6 [1]. The map is $\text{GF}(2)$ -linear, so agreement on the 32 unit-input vectors is a complete correctness check; counts and depths are relabeling-invariant, so no comparison below depends on convention.

The s-XOR (in-place) model is comparable: SSA-unrolling a k -instruction s-XOR program yields exactly k free-standing gates. Multi-input XOR gates, gate-equivalent area and quantum CNOT counts are different cost functions: not comparable, not claimed against.

2 Results

Depth 3 is the known minimum depth — an output depending on w inputs needs depth $\geq \lceil \log_2 w \rceil$, and MixColumns has outputs of weight 7 [2] — so the depth-3 circuit’s contribution is its count. The 89 at depth 5 is shallower than any published circuit of fewer than 94 gates whose depth is stated, and stays on the frontier: the published sub-89 point, Jean’s 88 [9], measures depth 7 on our transcription — that note states no depth, exactly as [8] does not.

The 88 at depth 7 *ties* Jean’s point rather than improving it: Jean’s note was posted 2026-07-23 and has priority. Ours was found 2026-07-26 on a fully logged own lineage — from-scratch 97 at depth 3 (235s), a 21 h ladder to 89 at depth 6 (11.0 h), one reroute to 89 at depth 5 (592s), ρ^2 -symmetrization to a 94 at depth 5, then 32.9 min of plateau walking

Depth	Published best	Model	This work	Relation
3	99 [2]	XOR2	97	−2 (depth 3 is the minimum depth)
4	97 [3]	XOR2	92	−5
5	94 [3]	XOR2	89	−5
6	92 [4, 5]	XOR2 / s-XOR	—	dominated by 89 @ 5
7 [‡]	88 [9]	XOR2	88	ties it; independent, 61/88 masks shared
8	—	XOR2	(88) [†]	derived from [9]; dominated by 88 @ 7
9 [‡]	89 [8]	g-XOR	—	off the frontier

Table 1: The published depth–count landscape in comparable models, and this work. [†]Derived and non-frontier: listed for completeness, claimed as nothing. [‡]*Neither* [9] *nor* [8] states a depth: the 7 and the 9 are both our own measurements of our own transcriptions, on the same footing. Jean’s 88 [9] supersedes the earlier 91 of [6, 7] at depth 7; the depth-3 lineage is 105 → 103 → 102 → 99 ([2], Table 3). Listings: Appendices A–E.

— with no imported circuit in the chain, and is a different circuit: 61 of 88 masks shared, Jaccard 0.530. The fifth circuit, 88 at depth 8 (Appendix E), is derived work and dominated: its seed chain passes through Jean’s, ρ^2 -symmetrized and peeled to 95, orbit-walked to 92, then unioned with a 91 of our own lineage before the descent.

3 Method

A circuit is represented as a *value set*: masks over the 32 inputs, valid iff every mask is the XOR of two already-available masks and all 32 output masks are present; gate count is the set size. Deletion becomes a set operation, circuits compare by mask overlap regardless of gate order, and a neighbourhood — remove k masks, add at most $k - 1$ — becomes finite and enumerable (Section 4).

Good circuits at this size *look* locally efficient — every gate reads as an optimal step — and that is what makes them hard to improve: they sit at local optima, so smaller circuits are separated from them by locally *suboptimal* steps. (This was true of the naive construction we started from and of every circuit we produced up to the 89; we make no claim about how anyone else’s circuit was built.) **Remove-1** drops a non-output mask if the set stays realizable. The **neutral swap** removes one mask and adds one repair mask at unchanged size: with $A = \text{closure}(S')$ and $\text{stuck} = (S' \cup T) \setminus A$, every single-mask repair lies in $C = \{u \oplus a : u \in \text{stuck}, a \in A\} \cap P_2(A)$ (P_2 = pair sums), so enumerating C *decides* the move — measured $|C| \approx 7$, cheaper than the earlier engine’s 24–40 random candidates, which missed 81–82% of existing repairs. **Remove-2-add-1** remains a diversifier, though at the 88/89 frontier it is provably empty (Section 4). **Destroy-and-rebuild** removes a group and re-synthesizes from a pool, accepting rebuilds no larger, or larger on an annealing schedule with reheat; destroying a *connected cone* of the DAG leaves a hole worth re-planning, where k unrelated masks are usually just re-derived, and near-miss rebuilds are peeled before judgement. A fixpoint over minimum build depths rejects cap violations and emits gate lists in depth order.

Workers run one engine at one depth cap in fixed-length chunks, alternating walk and rebuild, and a coordinator re-offers each the best circuit feasible at its cap (the reseeding ladder). A worker accepts fewer gates *or equal gates at strictly lower depth* — a Pareto tie-break, and not a cosmetic one: 89 at depth 6 became 89 at depth 5 through it, and both 88s, first seen at depths 11 and 10, reached depths 7 and 8 seconds later. Every candidate is verified against MixColumns rebuilt from $\text{GF}(2^8)$ before acceptance. The depth-3 circuit comes from a separate constructor: each output is modelled as $A \oplus B$, A and B partitioning its bits into shared, refcount-costed depth- ≤ 2 parts, annealed by iterated local search, so

depth ≤ 3 holds by construction.

The v2 engine. Version 1’s circuits came from an engine whose kernel was then profiled and rewritten [10]: level-BFS depth relaxation ($17.5\times$ on destroy-and-rebuild, bit-identical trajectories); a worklist closure with incremental removal queries, and removal of a bug that re-verified unchanged circuits every plateau iteration — 78 % of walk runtime ($21.9\times$ on the walk); victim repooling ($\sim 34\times$ accepted moves/s); cone destroys ($\sim 12\times$); exact repair ($\sim 7\times$ plateau mobility). The walk now runs at 480–640 iterations/s against 70–110. What found both 88s was not a cleverer move but a cheap plateau walked at volume: every distinct equal-best mask set is harvested.

4 Exact local certificates, and structure

To say what is *not* there we decide the neighbourhood question: given a verified 88-gate circuit and a window of k removed masks, can at most $k - 1$ new masks — each buildable when added, with cascade unlocks between — restore all 32 targets? A “yes” is an 87-gate circuit; a completed “no” is a proof for that window. Completeness follows from a first-unlock case analysis: the first unlock must have the newest added mask as a parent, which pins each added mask to a finite, enumerable shape. Runs above the proved budget are recorded as undecided, never as irreducible. The procedure agrees with an independent brute force 25/25 and 12/12 at $k \leq 3$, and on 1 257 instances including 122 genuine negatives at $k = 4$.

- **47 canonical 88-gate circuits** — Jean’s 88, its 12 plateau siblings, and a third family’s anchor with 33 representatives — **have exhaustively empty $k \leq 3$ shells**: all 1 540 windows at $k = 2$ and all 27 720 at $k = 3$ each, none reducible. Hence *any 87-gate circuit differs from each of the 47 by at least 4 masks*. Stated plainly: all 47 lie in Jean’s-lineage families — families 1–2 are Jean’s circuit and its siblings, and the third family’s anchor is itself seeded through Jean’s circuit — so this is a rigidity statement about *that* neighbourhood, and the one circuit here independent of that lineage, our own 88 at depth 7, is exactly the least-certified: it is *not* among the 47, its exhaustive $k \leq 3$ sweep was never run, and it has only 9 exact $k = 4$ windows and 8 SAT cone windows. The best-certified shell and the independent shell are disjoint.
- **Population.** Harvesting mapped $\approx 139\,878$ distinct 88-gate mask sets, of which 105 801 are proven irreducible at $k = 2$. No 88-gate mask set of minimum build depth ≤ 6 has been seen in this harvesting — an observation, not a certified claim: no per-circuit depth census of the harvest is published. In total ≈ 165 million exact window decisions, zero timeouts, zero reducible windows; all five 89-gate seeds likewise have empty $k = 2$ shells.
- **Windowed SAT is evidence, not proof.** A sound windowed CNF encoding, solved by Kissat under two soundness-proved symmetry-breaking layers, returned UNSAT up to $k = 16$ on Jean’s 88 and $k = 15$ on the third-family anchor (whose whole $k \leq 12$ regime is UNSAT, 28/28), with **0 SAT anywhere** — but UNSAT there is relative to the encoding’s fixed slot order, and undecided windows are undecided.

None of this bounds 87 away globally; it is consistent with 88 being locally rigid, and it redirected the search: at 88 gates every move class small enough to enumerate is empty, so what remains is walking the equal-size plateau widely.

Symmetry. Byte rotation ρ (order 4) commutes with MixColumns, so the 32 targets form 8 ρ -orbits of size 4 and equivariant circuits can be searched in orbit space — but full ρ -symmetry costs about +19 gates. The half-rotation ρ^2 is where the structure is: in the elite

basin 79–81 % of a circuit’s masks have their ρ^2 -image also in the mask set, and the 12 ρ^2 -fixed masks the search settles on are exactly the classic $x_0 \oplus x_2 / x_1 \oplus x_3$ pairs, rediscovered rather than imposed. Jean’s 88, at 75 % (66 of 88), symmetrizes into a second, distinct basin. A ρ^2 -equivariant orbit engine produced two *exactly* ρ^2 -symmetric 90-gate circuits (depths 9 and 7), improving the best exactly symmetric circuit we know of from 94 to 90; both are certified locally optimal in orbit space, and both are derived work, each seed union containing a circuit symmetrized from Jean’s 88. Symmetry is also a seeding device: the exactly ρ^2 -symmetric 94 at depth 5 — 82 of its 94 masks shared with the 89 at depth 5 — costs +5 over that 89 and is the seed from which the 88 at depth 7 was found.

5 Verification, artifacts, and reproduction

Every circuit here is checkable from the note alone: the appendices list every gate, Section 1 defines the target map, and because MixColumns is linear, replaying a listing on the 32 unit-input vectors against Eq. 5.6 [1] is a complete correctness check. Each appendix also states its circuit’s canonical SHA-256 over the UTF-8 bytes of `{"inputCount":32,"gates":[...]}`.

The artifacts automate that. Circuits and verifiers are at <https://github.com/Joe-b-20/aes-mixcolumns-xor-circuits>, archived at concept DOI 10.5281/zenodo.21299092; `python3 verify_all.py` runs `verify.py`, which rebuilds the target map from the $\text{GF}(2^8)$ definition so the verifier *is* the specification, a clean-room verifier with adversarial rejection tests, and Verilog testbenches, all under continuous integration. The method, pipeline, untouched run archives with each run’s exact code, certificate summaries and the credited transcriptions of the imported circuits [9, 8] are published separately [10], one reproduction command per result. Measured there on 2026-07-27 with the shipped v2 engine, and reported as measurements rather than promises: the from-scratch 97 at depth 3 in 81 s on one core; the 89 at depth 5 in 19 s and 22 s from shipped seeds, against 592 s for the archived 2026-07-14 run on the v1 engine; the 88 at depth 7 in 19.4 and 31.0 minutes in two single-worker re-runs from the ρ^2 -symmetric 94, against 32.9 min in the archived ten-worker run; the 92 at depth 4 a from-scratch cascade of hours. The re-runs are re-runs, not independent confirmations.

Implementation and execution used LLM coding agents as programming tools, directed by the author, who designed the moves and acceptance rules. Concretely: the three frontier records of this note are reproduced by the published, dependency-free code with no AI system in the loop, whereas the v2 engine, the certificates of Section 4 and both 88-gate circuits came out of a 24-agent LLM campaign over 2026-07-26/27, again author-directed. That distinction changes nothing about correctness: every circuit any of it produced is machine-verified against MixColumns rebuilt from $\text{GF}(2^8)$, and the method is fully specified and reproducible from the accompanying code either way.

6 Scope and claims

- **Not optimality.** Minimum 2-input-XOR circuit size (the Shortest Linear Program problem) is NP-hard; no count here is proven minimal, and the certificates of Section 4 are local.
- **Comparisons.** All counts are for the model and convention of Section 1; the baselines [2, 3, 4, 5, 6, 7, 8, 9] were source-checked, and a source-by-source audit, with a dated Corrections section, is in the artifact repository’s `PRIOR_ART.md`. We claim the smallest counts at depths 3, 4 and 5, and no improvement at unconstrained depth: the published floor is 88 and stays 88.

- **Priority.** The 88 at depth 7 ties [9], posted first; the 88 at depth 8 is derived from that circuit and dominated. Neither is a record claim. The depths quoted for [9] and [8] — 7 and 9 — are our measurements of our own transcriptions; neither paper states a depth.
- **Earlier results.** This note supersedes the project’s 98 at depth 3, 91 at depth 6 and 89 at depth 10, kept in the artifact repository.

Version history. **v2**, 27 July 2026: adds the 88 at depth 7 and its derived depth-8 companion, the exact neighbourhood certificates, the ρ^2 symmetry results, the v2 engine measurements and dated reproduction times; no claim of v1 is withdrawn. **v1**, July 2026 (the 97, 92 and 89 only), remains available in the repository history and in the Zenodo version archive under the concept DOI 10.5281/zenodo.21299092.

References

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- [10] J. Bachir, *The method: value-set SLP search, plateau walking, and exact local certificates* (METHODS.md, evidence/RESULTS.md, reproduce/), slp-plateau-search v2.0.0, 2026. <https://github.com/Joe-b-20/slp-plateau-search>

A Circuit mixcolumns_97gates_depth3 (97 gates, depth 3)

Signals $s_0, \dots, s_{31}$ are the inputs; gate k produces s_{32+k} . Canonical gate-list SHA-256 (see Section 5):

e4c4e678c4b7c22d889c6e060b43859c111b82b953d6d9b43c49c079c0545b56

s32 = s0 $\oplus$ s8	s57 = s7 $\oplus$ s31	s82 = s51 $\oplus$ s41	s107 = s75 $\oplus$ s18
s33 = s5 $\oplus$ s13	s58 = s23 $\oplus$ s31	s83 = s53 $\oplus$ s3	s108 = s80 $\oplus$ s66
s34 = s6 $\oplus$ s14	s59 = s33 $\oplus$ s14	s84 = s52 $\oplus$ s40	s109 = s84 $\oplus$ s66
s35 = s7 $\oplus$ s15	s60 = s35 $\oplus$ s17	s85 = s52 $\oplus$ s21	s110 = s62 $\oplus$ s55
s36 = s8 $\oplus$ s16	s61 = s39 $\oplus$ s10	s86 = s42 $\oplus$ s54	s111 = s86 $\oplus$ s89
s37 = s9 $\oplus$ s17	s62 = s41 $\oplus$ s5	s87 = s53 $\oplus$ s54	s112 = s93 $\oplus$ s64
s38 = s10 $\oplus$ s18	s63 = s35 $\oplus$ s40	s88 = s43 $\oplus$ s56	s113 = s67 $\oplus$ s58
s39 = s11 $\oplus$ s19	s64 = s42 $\oplus$ s7	s89 = s55 $\oplus$ s30	s114 = s94 $\oplus$ s72
s40 = s11 $\oplus$ s20	s65 = s36 $\oplus$ s43	s90 = s57 $\oplus$ s0	s115 = s75 $\oplus$ s71
s41 = s12 $\oplus$ s20	s66 = s43 $\oplus$ s19	s91 = s57 $\oplus$ s3	s116 = s95 $\oplus$ s79
s42 = s6 $\oplus$ s22	s67 = s32 $\oplus$ s24	s92 = s57 $\oplus$ s15	s117 = s96 $\oplus$ s81
s43 = s15 $\oplus$ s23	s68 = s36 $\oplus$ s24	s93 = s34 $\oplus$ s58	s118 = s87 $\oplus$ s62
s44 = s1 $\oplus$ s24	s69 = s44 $\oplus$ s37	s94 = s44 $\oplus$ s58	s119 = s89 $\oplus$ s34
s45 = s1 $\oplus$ s25	s70 = s32 $\oplus$ s46	s95 = s48 $\oplus$ s58	s120 = s92 $\oplus$ s56
s46 = s9 $\oplus$ s25	s71 = s46 $\oplus$ s10	s96 = s53 $\oplus$ s58	s121 = s90 $\oplus$ s36
s47 = s2 $\oplus$ s26	s72 = s46 $\oplus$ s16	s97 = s68 $\oplus$ s35	s122 = s69 $\oplus$ s90
s48 = s11 $\oplus$ s26	s73 = s45 $\oplus$ s17	s98 = s70 $\oplus$ s60	s123 = s74 $\oplus$ s2
s49 = s2 $\oplus$ s27	s74 = s45 $\oplus$ s38	s99 = s74 $\oplus$ s77	s124 = s76 $\oplus$ s91
s50 = s3 $\oplus$ s27	s75 = s47 $\oplus$ s37	s100 = s78 $\oplus$ s61	s125 = s82 $\oplus$ s91
s51 = s4 $\oplus$ s27	s76 = s47 $\oplus$ s39	s101 = s63 $\oplus$ s83	s126 = s85 $\oplus$ s33
s52 = s4 $\oplus$ s28	s77 = s46 $\oplus$ s26	s102 = s87 $\oplus$ s85	s127 = s86 $\oplus$ s59
s53 = s12 $\oplus$ s28	s78 = s49 $\oplus$ s35	s103 = s59 $\oplus$ s56	s128 = s88 $\oplus$ s64
s54 = s13 $\oplus$ s29	s79 = s50 $\oplus$ s18	s104 = s93 $\oplus$ s15	
s55 = s21 $\oplus$ s29	s80 = s50 $\oplus$ s38	s105 = s65 $\oplus$ s67	
s56 = s22 $\oplus$ s30	s81 = s51 $\oplus$ s19	s106 = s65 $\oplus$ s73	

Output signals (output bit j is carried by the listed signal, $j = 0, \dots, 31$):

[97, 98, 99, 100, 101, 102, 103, 104, 105, 106, 107, 108, 109, 110, 111, 112, 113, 114, 115, 116, 117, 118, 119, 120, 121, 122, 123, 124, 125, 126, 127, 128]

B Circuit mixcolumns_92gates_depth4 (92 gates, depth 4)

Signals $s_0, \dots, s_{31}$ are the inputs; gate k produces s_{32+k} . Canonical gate-list SHA-256 (see Section 5):

e79bfccf5322fe8933242b47b53e94e73163615c70d5c45c9721e27ac86fd97d

s32 = s10 $\oplus$ s26	s56 = s15 $\oplus$ s31	s80 = s45 $\oplus$ s49	s104 = s63 $\oplus$ s49
s33 = s12 $\oplus$ s20	s57 = s43 $\oplus$ s44	s81 = s21 $\oplus$ s33	s105 = s64 $\oplus$ s50
s34 = s14 $\oplus$ s30	s58 = s34 $\oplus$ s53	s82 = s67 $\oplus$ s81	s106 = s64 $\oplus$ s69
s35 = s6 $\oplus$ s14	s59 = s5 $\oplus$ s54	s83 = s57 $\oplus$ s66	s107 = s23 $\oplus$ s75
s36 = s11 $\oplus$ s28	s60 = s0 $\oplus$ s37	s84 = s58 $\oplus$ s77	s108 = s71 $\oplus$ s80
s37 = s8 $\oplus$ s16	s61 = s38 $\oplus$ s52	s85 = s42 $\oplus$ s81	s109 = s46 $\oplus$ s72
s38 = s7 $\oplus$ s31	s62 = s6 $\oplus$ s40	s86 = s59 $\oplus$ s67	s110 = s102 $\oplus$ s66
s39 = s17 $\oplus$ s25	s63 = s32 $\oplus$ s56	s87 = s59 $\oplus$ s40	s111 = s18 $\oplus$ s101
s40 = s13 $\oplus$ s21	s64 = s36 $\oplus$ s56	s88 = s10 $\oplus$ s73	s112 = s89 $\oplus$ s78
s41 = s2 $\oplus$ s26	s65 = s37 $\oplus$ s56	s89 = s45 $\oplus$ s79	s113 = s63 $\oplus$ s108
s42 = s5 $\oplus$ s29	s66 = s1 $\oplus$ s48	s90 = s23 $\oplus$ s68	s114 = s108 $\oplus$ s55
s43 = s16 $\oplus$ s24	s67 = s50 $\oplus$ s53	s91 = s62 $\oplus$ s52	s115 = s93 $\oplus$ s65
s44 = s9 $\oplus$ s25	s68 = s35 $\oplus$ s56	s92 = s58 $\oplus$ s62	s116 = s47 $\oplus$ s109
s45 = s18 $\oplus$ s26	s69 = s3 $\oplus$ s38	s93 = s60 $\oplus$ s76	s117 = s52 $\oplus$ s107
s46 = s0 $\oplus$ s17	s70 = s51 $\oplus$ s54	s94 = s35 $\oplus$ s77	s118 = s96 $\oplus$ s104
s47 = s1 $\oplus$ s9	s71 = s19 $\oplus$ s48	s95 = s15 $\oplus$ s61	s119 = s33 $\oplus$ s106
s48 = s23 $\oplus$ s31	s72 = s24 $\oplus$ s38	s96 = s69 $\oplus$ s74	s120 = s71 $\oplus$ s105
s49 = s3 $\oplus$ s27	s73 = s39 $\oplus$ s41	s97 = s65 $\oplus$ s72	s121 = s103 $\oplus$ s51
s50 = s4 $\oplus$ s20	s74 = s41 $\oplus$ s55	s98 = s38 $\oplus$ s60	s122 = s97 $\oplus$ s100
s51 = s12 $\oplus$ s27	s75 = s34 $\oplus$ s38	s99 = s70 $\oplus$ s71	s123 = s90 $\oplus$ s75
s52 = s22 $\oplus$ s30	s76 = s43 $\oplus$ s48	s100 = s57 $\oplus$ s46	
s53 = s13 $\oplus$ s29	s77 = s22 $\oplus$ s42	s101 = s39 $\oplus$ s78	
s54 = s4 $\oplus$ s28	s78 = s41 $\oplus$ s44	s102 = s39 $\oplus$ s65	
s55 = s11 $\oplus$ s19	s79 = s10 $\oplus$ s47	s103 = s69 $\oplus$ s50	

Output signals (output bit j is carried by the listed signal, $j = 0, \dots, 31$):

[97, 122, 89, 118, 119, 82, 84, 90, 115, 110, 111, 113, 120, 85, 91, 117, 93, 83, 88, 114, 99, 86, 92, 95, 98, 116, 112, 96, 121, 87, 94, 123]

C Circuit mixcolumns_89gates_depth5 (89 gates, depth 5)

Signals $s_0, \dots, s_{31}$ are the inputs; gate k produces s_{32+k} . Canonical gate-list SHA-256 (see Section 5):

c3b972af282d25e8313a510f5cf840472401c6268bd28cda889593fc3379f064

s32 = s9 $\oplus$ s17	s55 = s21 $\oplus$ s39	s78 = s55 $\oplus$ s66	s101 = s60 $\oplus$ s50
s33 = s19 $\oplus$ s27	s56 = s5 $\oplus$ s46	s79 = s37 $\oplus$ s57	s102 = s39 $\oplus$ s101
s34 = s1 $\oplus$ s25	s57 = s14 $\oplus$ s42	s80 = s35 $\oplus$ s56	s103 = s34 $\oplus$ s87
s35 = s21 $\oplus$ s29	s58 = s15 $\oplus$ s37	s81 = s56 $\oplus$ s66	s104 = s59 $\oplus$ s89
s36 = s12 $\oplus$ s28	s59 = s11 $\oplus$ s33	s82 = s55 $\oplus$ s42	s105 = s76 $\oplus$ s70
s37 = s22 $\oplus$ s30	s60 = s11 $\oplus$ s43	s83 = s70 $\oplus$ s72	s106 = s92 $\oplus$ s65
s38 = s3 $\oplus$ s27	s61 = s0 $\oplus$ s52	s84 = s43 $\oplus$ s71	s107 = s96 $\oplus$ s98
s39 = s4 $\oplus$ s28	s62 = s33 $\oplus$ s45	s85 = s54 $\oplus$ s67	s108 = s36 $\oplus$ s93
s40 = s9 $\oplus$ s25	s63 = s38 $\oplus$ s52	s86 = s54 $\oplus$ s49	s109 = s95 $\oplus$ s48
s41 = s13 $\oplus$ s29	s64 = s43 $\oplus$ s51	s87 = s17 $\oplus$ s64	s110 = s1 $\oplus$ s92
s42 = s5 $\oplus$ s13	s65 = s8 $\oplus$ s45	s88 = s57 $\oplus$ s67	s111 = s99 $\oplus$ s46
s43 = s15 $\oplus$ s23	s66 = s36 $\oplus$ s41	s89 = s60 $\oplus$ s47	s112 = s98 $\oplus$ s75
s44 = s6 $\oplus$ s22	s67 = s41 $\oplus$ s44	s90 = s62 $\oplus$ s53	s113 = s89 $\oplus$ s90
s45 = s7 $\oplus$ s23	s68 = s34 $\oplus$ s53	s91 = s71 $\oplus$ s74	s114 = s108 $\oplus$ s99
s46 = s12 $\oplus$ s20	s69 = s24 $\oplus$ s45	s92 = s24 $\oplus$ s61	s115 = s110 $\oplus$ s112
s47 = s10 $\oplus$ s18	s70 = s40 $\oplus$ s47	s93 = s38 $\oplus$ s62	s116 = s3 $\oplus$ s104
s48 = s2 $\oplus$ s26	s71 = s31 $\oplus$ s49	s94 = s58 $\oplus$ s52	s117 = s90 $\oplus$ s109
s49 = s6 $\oplus$ s14	s72 = s26 $\oplus$ s34	s95 = s59 $\oplus$ s63	s118 = s108 $\oplus$ s101
s50 = s19 $\oplus$ s20	s73 = s32 $\oplus$ s48	s96 = s64 $\oplus$ s69	s119 = s87 $\oplus$ s112
s51 = s8 $\oplus$ s16	s74 = s44 $\oplus$ s45	s97 = s61 $\oplus$ s51	s120 = s32 $\oplus$ s110
s52 = s7 $\oplus$ s31	s75 = s40 $\oplus$ s51	s98 = s0 $\oplus$ s65	
s53 = s2 $\oplus$ s18	s76 = s18 $\oplus$ s73	s99 = s4 $\oplus$ s63	
s54 = s30 $\oplus$ s35	s77 = s10 $\oplus$ s68	s100 = s58 $\oplus$ s74	

Output signals (output bit j is carried by the listed signal, $j = 0, \dots, 31$):

[96, 119, 83, 113, 118, 78, 79, 84, 107, 103, 76, 116, 102, 80, 85, 91, 106, 115, 105, 117, 114, 81, 86, 94, 97, 120, 77, 109, 111, 82, 88, 100]

D Circuit mixcolumns_88gates_depth7 (88 gates, depth 7; ties the published floor)

Status: the same (depth, count) point as Jean [9], reached independently on this project's own lineage; 61 of 88 internal masks are shared. It ties that point and does not beat it, and Jean has priority.

Signals $s_0, \dots, s_{31}$ are the inputs; gate k produces s_{32+k} . Canonical gate-list SHA-256 (see Section 5):

76bac50ab0e4a7dd54578e77f57f6d08e96a8803aff72dba2c3f5fadaaa06ee1

s32 = s13 $\oplus$ s29	s43 = s5 $\oplus$ s13	s54 = s33 $\oplus$ s30	s65 = s43 $\oplus$ s63
s33 = s21 $\oplus$ s29	s44 = s15 $\oplus$ s31	s55 = s34 $\oplus$ s15	s66 = s44 $\oplus$ s45
s34 = s22 $\oplus$ s30	s45 = s10 $\oplus$ s26	s56 = s35 $\oplus$ s20	s67 = s45 $\oplus$ s58
s35 = s15 $\oplus$ s23	s46 = s6 $\oplus$ s22	s57 = s35 $\oplus$ s8	s68 = s46 $\oplus$ s49
s36 = s17 $\oplus$ s25	s47 = s6 $\oplus$ s14	s58 = s36 $\oplus$ s2	s69 = s47 $\oplus$ s54
s37 = s12 $\oplus$ s20	s48 = s11 $\oplus$ s19	s59 = s37 $\oplus$ s5	s70 = s47 $\oplus$ s31
s38 = s12 $\oplus$ s28	s49 = s7 $\oplus$ s23	s60 = s39 $\oplus$ s1	s71 = s48 $\oplus$ s56
s39 = s9 $\oplus$ s25	s50 = s7 $\oplus$ s31	s61 = s39 $\oplus$ s51	s72 = s49 $\oplus$ s19
s40 = s2 $\oplus$ s18	s51 = s18 $\oplus$ s26	s62 = s41 $\oplus$ s50	s73 = s49 $\oplus$ s16
s41 = s3 $\oplus$ s27	s52 = s32 $\oplus$ s46	s63 = s42 $\oplus$ s21	s74 = s50 $\oplus$ s24
s42 = s4 $\oplus$ s28	s53 = s32 $\oplus$ s38	s64 = s43 $\oplus$ s14	s75 = s50 $\oplus$ s55

s76 = s51 $\oplus$ s72	s88 = s64 $\oplus$ s34	s100 = s86 $\oplus$ s48	s112 = s104 $\oplus$ s87
s77 = s52 $\oplus$ s54	s89 = s66 $\oplus$ s86	s101 = s87 $\oplus$ s37	s113 = s104 $\oplus$ s71
s78 = s52 $\oplus$ s64	s90 = s67 $\oplus$ s85	s102 = s90 $\oplus$ s61	s114 = s106 $\oplus$ s108
s79 = s53 $\oplus$ s63	s91 = s68 $\oplus$ s70	s103 = s90 $\oplus$ s40	s115 = s107 $\oplus$ s74
s80 = s53 $\oplus$ s59	s92 = s70 $\oplus$ s35	s104 = s94 $\oplus$ s38	s116 = s107 $\oplus$ s110
s81 = s55 $\oplus$ s68	s93 = s71 $\oplus$ s42	s105 = s94 $\oplus$ s40	s117 = s107 $\oplus$ s57
s82 = s57 $\oplus$ s16	s94 = s72 $\oplus$ s3	s106 = s95 $\oplus$ s60	s118 = s111 $\oplus$ s41
s83 = s58 $\oplus$ s61	s95 = s73 $\oplus$ s74	s107 = s96 $\oplus$ s44	s119 = s111 $\oplus$ s66
s84 = s59 $\oplus$ s33	s96 = s73 $\oplus$ s0	s108 = s96 $\oplus$ s36	
s85 = s60 $\oplus$ s17	s97 = s82 $\oplus$ s9	s109 = s97 $\oplus$ s108	
s86 = s62 $\oplus$ s76	s98 = s82 $\oplus$ s24	s110 = s98 $\oplus$ s49	
s87 = s62 $\oplus$ s4	s99 = s85 $\oplus$ s97	s111 = s100 $\oplus$ s105	

Output signals (output bit j is carried by the listed signal, $j = 0, \dots, 31$):

[110, 109, 103, 119, 113, 79, 88, 92, 115, 99, 83, 89, 93, 84, 77, 91, 116, 106, 67, 100, 112, 80, 69, 75, 117, 114, 102, 118, 101, 65, 78, 81]

E Circuit mixcolumns_88gates_depth8 (88 gates, depth 8; derived, non-frontier)

Status: derived from published work, and not a frontier point. Its seed chain passes through Jean's published 88-gate circuit [9] (Section 2), and it is dominated by Appendix D (same count, greater depth). It is listed as a verified distinct construction, not as a claim.

Signals $s_0, \dots, s_{31}$ are the inputs; gate k produces s_{32+k} . Canonical gate-list SHA-256 (see Section 5):

3a23e2f2b2f3835cab74ba23800088bb03ab4ee3f076f2b9a826c78cd84f8ef1

s32 = s13 $\oplus$ s29	s55 = s35 $\oplus$ s8	s78 = s59 $\oplus$ s75	s101 = s95 $\oplus$ s20
s33 = s21 $\oplus$ s29	s56 = s38 $\oplus$ s26	s79 = s59 $\oplus$ s63	s102 = s96 $\oplus$ s16
s34 = s22 $\oplus$ s30	s57 = s39 $\oplus$ s50	s80 = s60 $\oplus$ s34	s103 = s96 $\oplus$ s8
s35 = s15 $\oplus$ s23	s58 = s40 $\oplus$ s21	s81 = s61 $\oplus$ s41	s104 = s97 $\oplus$ s36
s36 = s9 $\oplus$ s17	s59 = s42 $\oplus$ s45	s82 = s63 $\oplus$ s76	s105 = s100 $\oplus$ s89
s37 = s12 $\oplus$ s28	s60 = s43 $\oplus$ s14	s83 = s64 $\oplus$ s66	s106 = s100 $\oplus$ s39
s38 = s2 $\oplus$ s18	s61 = s43 $\oplus$ s58	s84 = s66 $\oplus$ s35	s107 = s101 $\oplus$ s94
s39 = s3 $\oplus$ s27	s62 = s45 $\oplus$ s19	s85 = s67 $\oplus$ s19	s108 = s101 $\oplus$ s37
s40 = s4 $\oplus$ s28	s63 = s46 $\oplus$ s26	s86 = s67 $\oplus$ s11	s109 = s102 $\oplus$ s44
s41 = s4 $\oplus$ s20	s64 = s47 $\oplus$ s49	s87 = s68 $\oplus$ s0	s110 = s102 $\oplus$ s68
s42 = s9 $\oplus$ s25	s65 = s48 $\oplus$ s53	s88 = s74 $\oplus$ s17	s111 = s106 $\oplus$ s86
s43 = s5 $\oplus$ s13	s66 = s48 $\oplus$ s31	s89 = s76 $\oplus$ s44	s112 = s106 $\oplus$ s38
s44 = s15 $\oplus$ s31	s67 = s49 $\oplus$ s3	s90 = s77 $\oplus$ s41	s113 = s109 $\oplus$ s68
s45 = s10 $\oplus$ s18	s68 = s50 $\oplus$ s24	s91 = s81 $\oplus$ s33	s114 = s110 $\oplus$ s42
s46 = s1 $\oplus$ s25	s69 = s50 $\oplus$ s54	s92 = s81 $\oplus$ s51	s115 = s112 $\oplus$ s93
s47 = s6 $\oplus$ s22	s70 = s51 $\oplus$ s58	s93 = s85 $\oplus$ s89	s116 = s113 $\oplus$ s103
s48 = s6 $\oplus$ s14	s71 = s52 $\oplus$ s53	s94 = s85 $\oplus$ s40	s117 = s113 $\oplus$ s55
s49 = s7 $\oplus$ s23	s72 = s52 $\oplus$ s60	s95 = s86 $\oplus$ s35	s118 = s114 $\oplus$ s97
s50 = s7 $\oplus$ s31	s73 = s54 $\oplus$ s64	s96 = s87 $\oplus$ s49	s119 = s114 $\oplus$ s88
s51 = s32 $\oplus$ s37	s74 = s55 $\oplus$ s16	s97 = s87 $\oplus$ s1	
s52 = s32 $\oplus$ s47	s75 = s56 $\oplus$ s36	s98 = s88 $\oplus$ s46	
s53 = s33 $\oplus$ s30	s76 = s56 $\oplus$ s10	s99 = s94 $\oplus$ s77	
s54 = s34 $\oplus$ s15	s77 = s57 $\oplus$ s12	s100 = s95 $\oplus$ s62	

Output signals (output bit j is carried by the listed signal, $j = 0, \dots, 31$):

[116, 119, 79, 112, 108, 70, 80, 84, 109, 98, 75, 111, 107, 92, 71, 83, 103, 118, 78, 115, 99, 91, 65, 69, 117, 104, 82, 105, 90, 61, 72, 73]